Lecture 10. Unsupervised Learning

Liqing Zhang & Shikui Tu

{zhang-lq, tushikui} @cs.sjtu.edu.cn

Shanghai Jiao Tong University

Unsupervised Learning

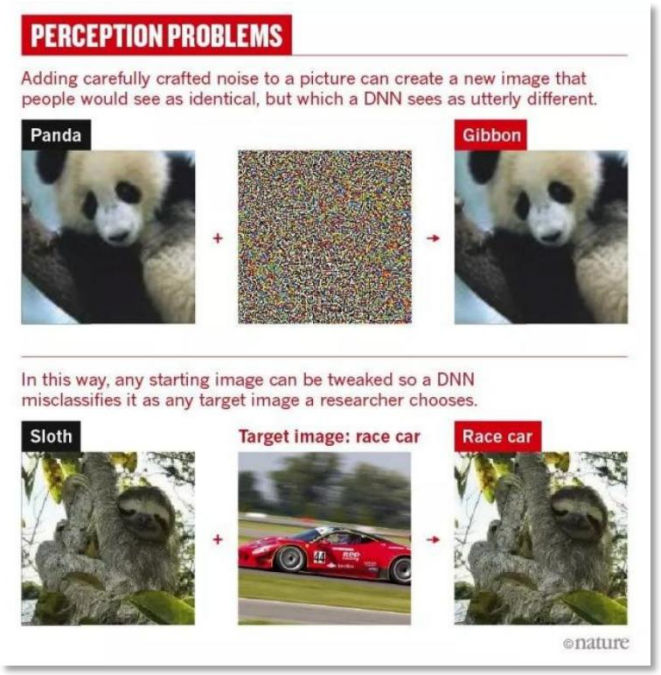
1. Introduction
2. Cluster Analysis
3. Principal Components, SVD
4. Non-negative Matrix Factorization
5. Independent Component Analysis
6. Multidimensional Scaling
7. Nonlinear Dimension Reduction
8. The Google PageRank Algorithm

Motivation

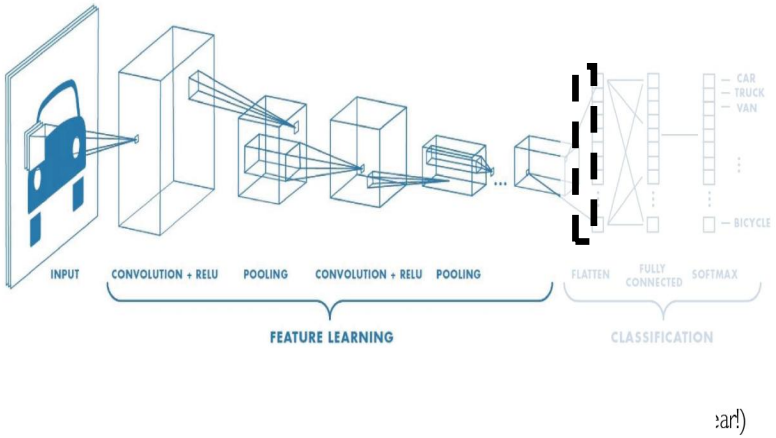
- Supervised Learning: Large scale of labeled data
- Unsupervised Learning:
  - Latent structures in data: linear or nonlinear
  - Dimension reduction
  - Relations in data
- Semi-supervised learning
- Reinforcement learning

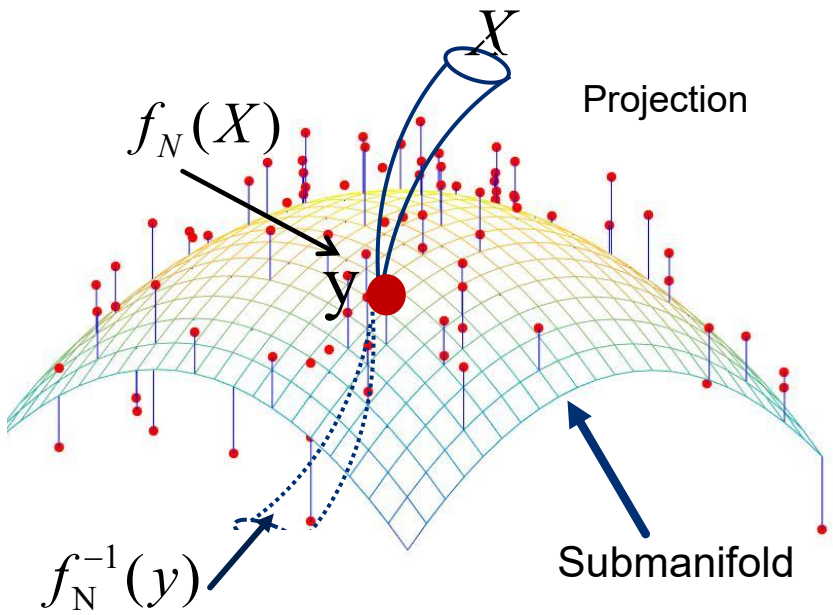
AI Challenges

- 维数灾难问题: CNN mapping images into embedding vectors, without expressing any semantic structures



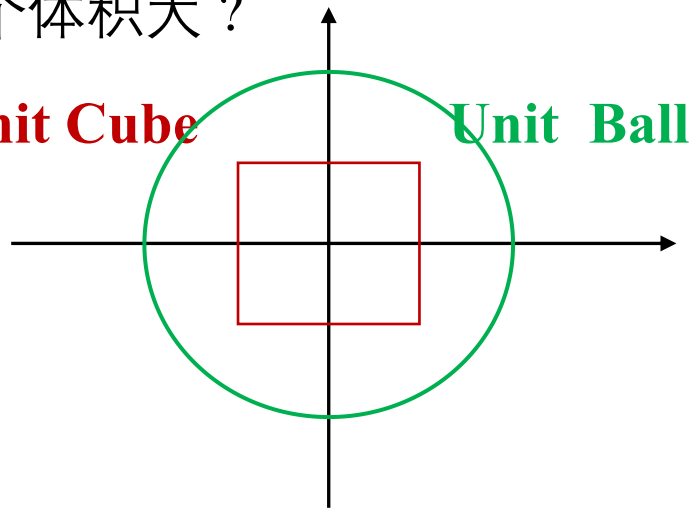
$X \in R^{M}$ $\quad f_N(X) \quad$ $y \in R^{N}$

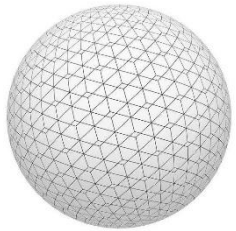




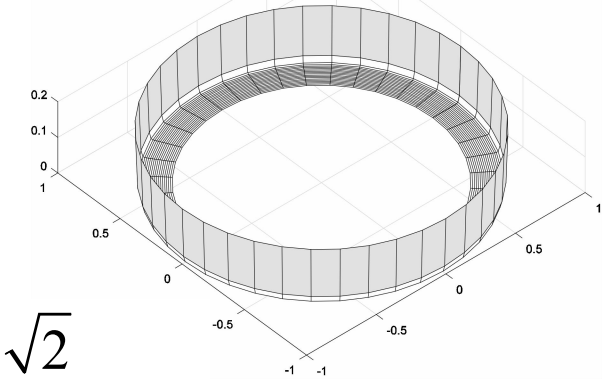
Data Distributions in High Dimensional Space

- **问题 1:** 高位空间中单位球和单位立方体哪个体积大？
-



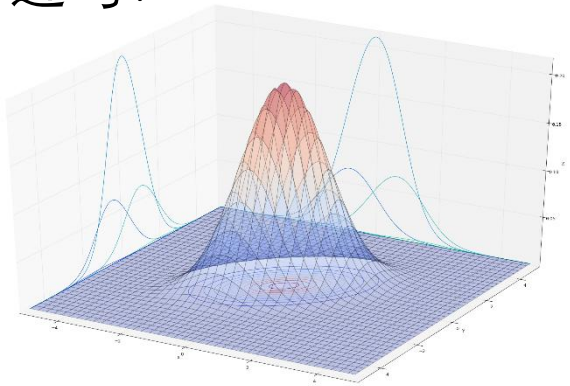


均匀分布

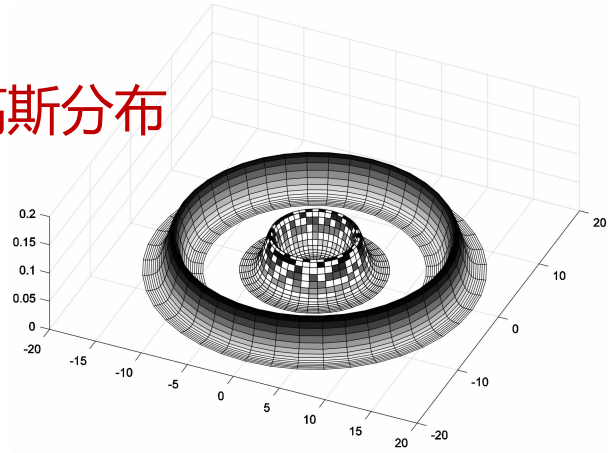


单位球内任意两点距离≈ $\sqrt{2}$

- **问题 2:** 高维空间中高斯分布的数据主要集中在均值附近吗？



高斯分布



内圈 n=20; 外圈 n=200, 半径 $r=\sqrt{n-1}$

1. Introduction

- Unsupervised learning or "learning without a teacher."
- A set of N observations

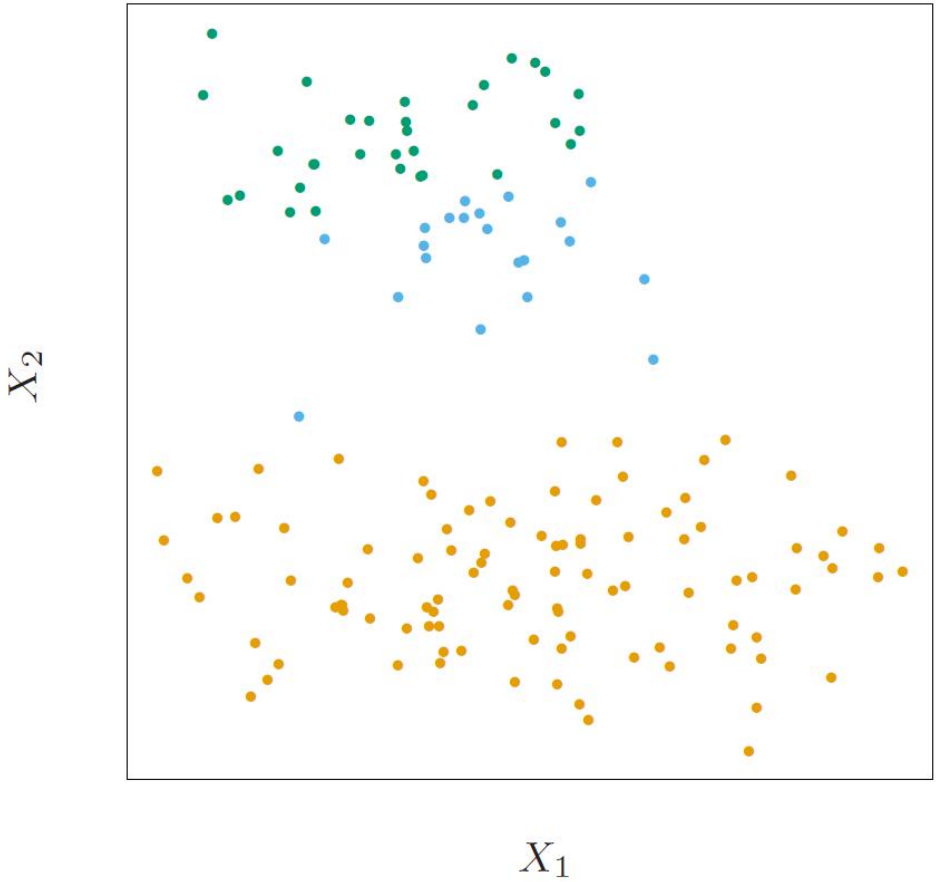
$$\{x_1, x_2, \cdots, x_N\}, \quad X \in R^p \sim p(X)$$

- In low-dimensional problems (say p ≤ 3), there are a variety of effective nonparametric methods for directly estimating the density $p(X)$ itself at all X-values
- In high-dimensional problems, one must settle for estimating rather crude global models, such as Gaussian mixtures or various simple descriptive statistics that characterize $p(X)$.

2. Cluster Analysis

- Cluster analysis, also called data segmentation, has a variety of goals.

All relate to grouping or segmenting a collection of objects into subsets or “clusters,” such that those within each cluster are more closely related to one another than objects assigned to different clusters.



2. Cluster Analysis

- Central to all of the goals of cluster analysis is the notion of the degree of similarity(相似值) (or dissimilarity（相异值）) between the individual objects being clustered.
- K-means clustering starts with guesses for three cluster centers.
  - for each data point, the closest cluster center (in Euclidean distance) is identified;
  - each cluster center is replaced by the coordinate-wise average of all data points that are closest to it.

Object Dissimilarity

- Dissimilarity $D(x_i, x_{i'})$ between two objects or observations

$$D(x_i, x_{i'}) = \sum_{j=1}^{p} w_j d(x_{ij}, x_{i'j}); \quad \sum_{j=1}^{p} w_j = 1$$

- The average object dissimilarity measure over all pairs of observations in the data set

$$\bar{D} = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{i'=1}^{N} D(x_i, x_{i'}) = \sum_{j=1}^{p} w_j \bar{d}_j,$$

$$\bar{d}_j = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{i'=1}^{N} d_j(x_{ij}, x_{i'j})$$

Combinatorial Algorithms

- Assign each observation to a group or cluster without regard to a probability model describing the data.

$$\{x_i\}_{i=1}^{N} \qquad x_i \Rightarrow C(i) = k, k = 1, \ldots, K$$

- The Loss function

$$W(C) = \frac{1}{2} \sum_{k=1}^{K} \sum_{C(i)=k} \sum_{C(i')=k} d(x_i, x_{i'})$$

- assigned to the same cluster tend to be close to one another.

Combinatorial Algorithms

- The *total* point scatter

$$\mathrm{T} = \frac{1}{2}\sum_{i=1}^{N}\sum_{i'=1}^{N} d(x_i, x_{i'}) = \frac{1}{2}\sum_{k=1}^{K}\sum_{C(i)=k}\left(\sum_{C(i')=k} d(x_i, x_{i'}) + \sum_{C(i')\neq k} d(x_i, x_{i'})\right)$$

$$= W(C) + B(C)$$

- Between-Cluster Scatter:

$$\mathrm{B}(C) = \frac{1}{2}\sum_{k=1}^{K}\sum_{C(i)=k}\sum_{C(i')\neq k} d(x_i, x_{i'})$$

- Minimizing W(C) is equivalent to *maximizing* B(C).

K-means

- The within-point scatter can be written as

$$W(C)=\frac{1}{2}\sum_{k=1}^{K}\sum_{C(i)=k}\sum_{C(i')=k}d(x_i,x_{i'})$$

$$=\frac{1}{2}\sum_{k=1}^{K}\sum_{C(i)=k}\sum_{C(i')=k}\|x_i-x_{i'}\|^2=\sum_{k=1}^{K}N_k\sum_{C(i)=k}\|x_i-\bar{x}_k\|^2$$

where N_k is the sample number of k-th class

Algorithm: K-means Clustering.

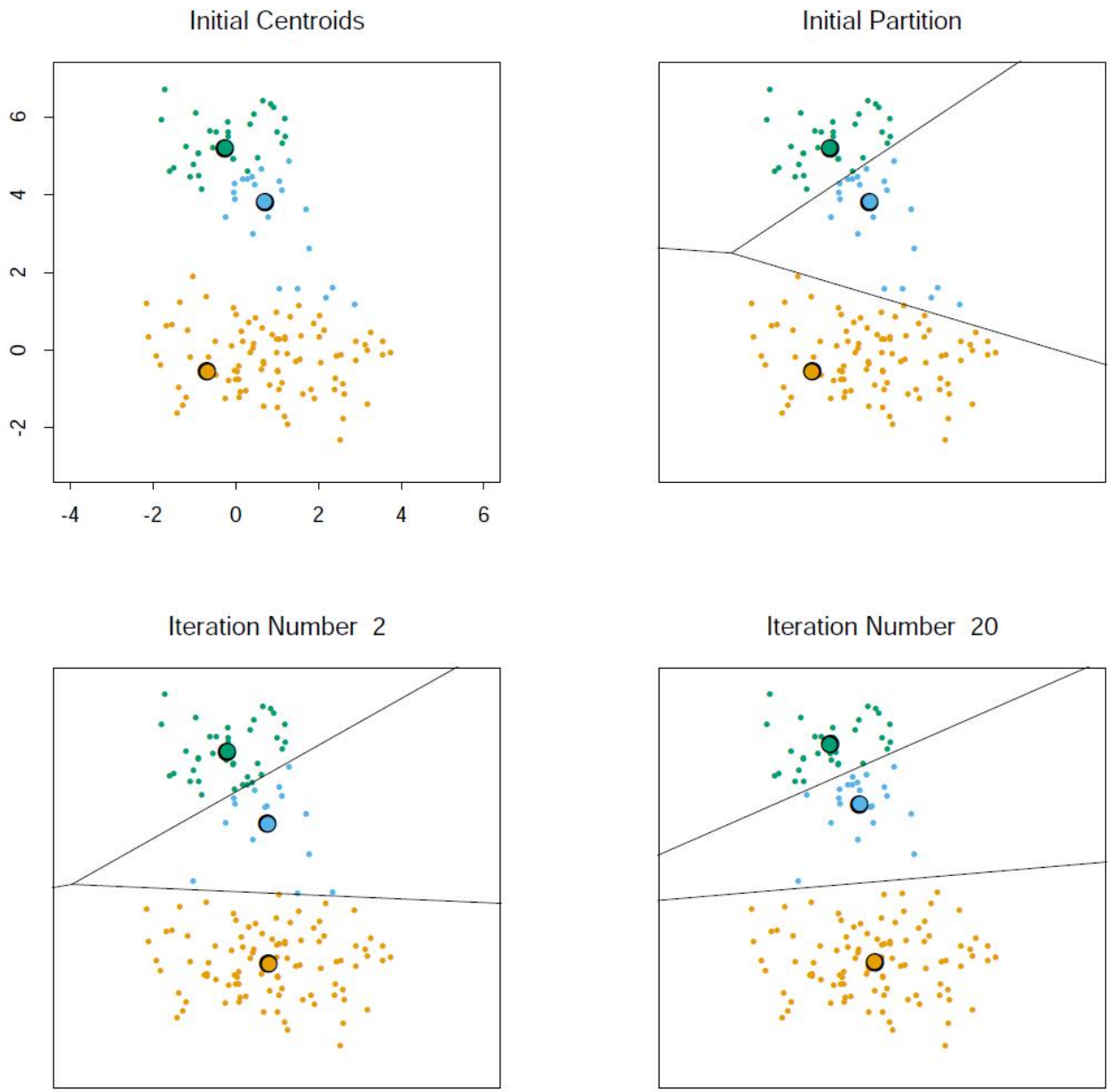
1. Initialize $C = \{C(i)\}_{i=1}^{K}$
2. For a given cluster assignment C, the total cluster variance is minimized with respect to $\{\mu_k\}_{k=1}^{K}$, yielding the means of the currently assigned clusters.

$$\{\mu_k\}_{k=1}^{K} = \arg\min_{\{\mu_k\}_{k=1}^{K}} \sum_{k=1}^{K} N_k \sum_{C(i)=k} \|x_i - \mu_k\|^2$$

3. Given a current set of means $\{\mu_k\}_{k=1}^{K}$, minimize the following cost function by assigning each observation to the closest (current) cluster mean

$$C(i) = \arg\min_k \|x_i - \mu_k\|^2$$

4. Repeat Steps 2 and 3 until the assignments do not change.


Initial Centroids
Initial Partition
Iteration Number 2
Iteration Number 20
-4
-2
0
2
4
6
-2
0
2
4
6

Unsupervised Learning

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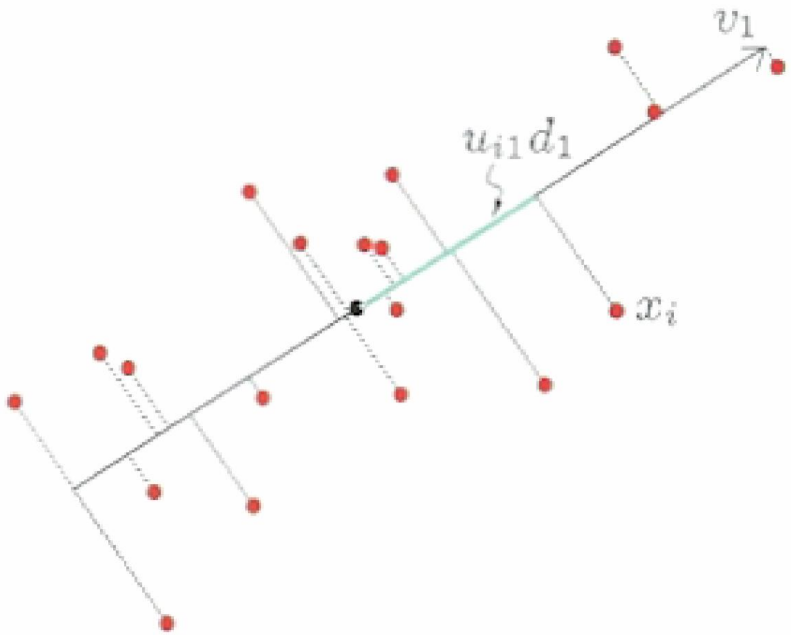
Principal Components (PCA)

Set of q directions towards which p-dimensional data are linearly projected

-> dimension reduction since $q \leq p$

Data : $x_1, x_2, \cdots, x_n$

Model : $f(\lambda) = (\mu + \lambda_i V_q)$



Model fitted by least squares

w.r.t. reconstruction error

$$\{\mu, \lambda_i, V_q\} = \text{var min}_{\mu,\lambda,V_q} \sum_{i=1}^{n} \left\| x_i - (\mu + \lambda_i V_q) \right\|^2$$

Principal Components (PCA)

Principal Components are computed from matrix operations

1. Given V_q, solve the optimal problem

$$\hat{\mu} = \bar{x}, \quad \hat{\lambda}_i = V_q^T (x_i - \bar{x}).$$

2. Substitute them into original cost function

$$V_q = \text{var}\min_{V_q} \sum_{i=1}^{n} \left\| (x_i - \bar{x}) - V_q V_q^T (x_i - \bar{x}) \right\|^2$$

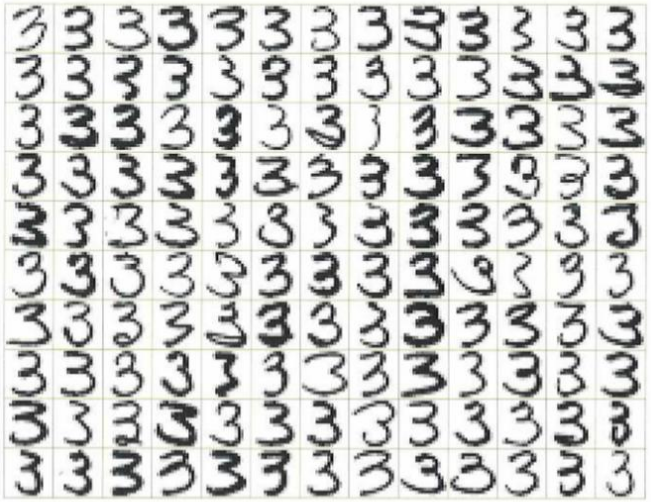
Data are usually centered, and $V_q V_q^T = H_q$ is the projection matrix. Solution V_q are the q first columns of V obtained from

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^T = \mathbf{S}\mathbf{A}^T; \ \mathbf{U} \ is \ \mathrm{N} \times \mathrm{p} \text{ orthognal matrix } \mathrm{U}^T\mathrm{U}=\mathrm{I}$$

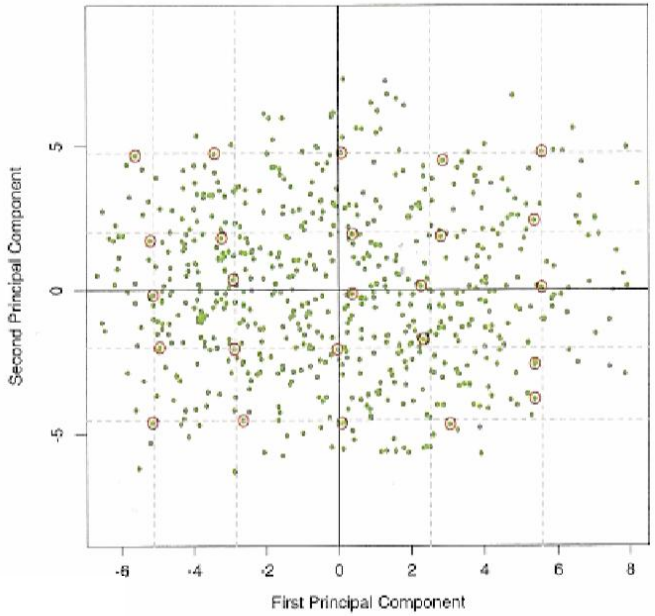
Principal Components (PCA)

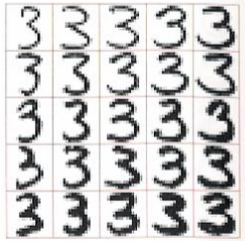
- PCA can be used for handwritten digits

Data in $\mathbb{R}^{256}$



2-D projections





$$\hat{f}(\lambda) = \bar{x} + \lambda_1 v_1 + \lambda_2 v_2$$

$$= \boxed{3} + \lambda_1 \cdot \boxed{3} + \lambda_2 \cdot \boxed{3}.$$

$$\hat{f}(\lambda) = \bar{x} + \lambda_1 v_1 + \lambda_2 v_2$$

SVD (Singular Value Decomposition)

- Definition:
  - **Singular value decomposition** is a method of decomposing a matrix into the product of three matrices. For any $m \times n$ matrix $\boldsymbol{A}$, there exist an $m \times m$ orthogonal matrix $\boldsymbol{U}$, an $n \times n$ orthogonal matrix $\boldsymbol{V}$, and an $m \times n$ diagonal matrix $\boldsymbol{D}$ such that

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T$$

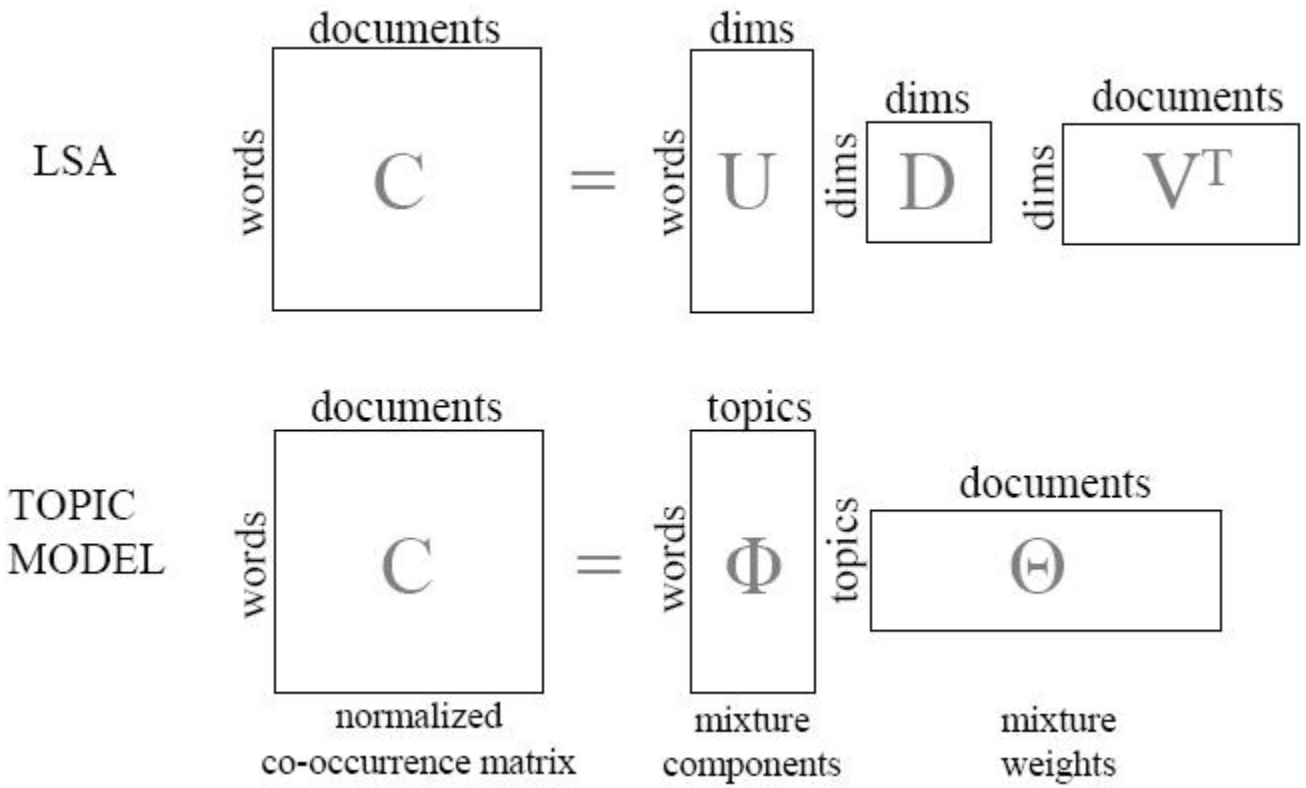
- Importance: It has wide applications in many fields such as data compression, image recognition, recommendation systems, and principal component analysis. It is an effective means of handling large-scale data and complex matrix operations.

SVD (Singular Value Decomposition)

- Interpretation:

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T$$

- Data representation



SVD (Singular Value Decomposition)

Proof outline: $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{V}\boldsymbol{D}\boldsymbol{V}^T$, $\quad (\boldsymbol{A}\boldsymbol{V})^T\boldsymbol{A}\boldsymbol{V} = \boldsymbol{D} = \left[diag(\lambda_1^2, \lambda_2^2, \cdots, \lambda_n^2)\right]$,

Denote $\boldsymbol{P} = \boldsymbol{A}\boldsymbol{V}$, then

$$\boldsymbol{P}^T\boldsymbol{P} = \boldsymbol{D}; \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r\,;\ \lambda_k = 0\ ,\mathrm{k} > \mathrm{r}$$

Define the orthogonal matrix $\boldsymbol{P} = [\boldsymbol{p}_1, \boldsymbol{p}_2, \cdots, \boldsymbol{p}_n]$, $\quad \boldsymbol{u}_\mathrm{j} = \boldsymbol{p}_j / \lambda_j$, $\mathrm{j} = 1,2,\cdots,\mathrm{r}$

$$\boldsymbol{u}_i^T\boldsymbol{u}_\mathrm{j} = \delta_\mathrm{ij}\ ,\mathrm{i},\mathrm{j} = 1,2,\cdots,\mathrm{r}$$

- Construct the rest vectors $\boldsymbol{u}_\mathrm{j},\ j > r$ from the null space of $\mathbf{A}^T$ such that $\{\boldsymbol{u}_j\}_{j=1}^m$, are orthogonal.

$$\boldsymbol{U}_r\boldsymbol{A}\boldsymbol{V} = \boldsymbol{U}_r\boldsymbol{P} = \boldsymbol{\Lambda}_r = \mathrm{diag}(\lambda_1, \cdots \lambda_\mathrm{r})$$

$$\boldsymbol{u}_i^T\boldsymbol{A} = 0, \quad i = r, \cdots, m$$

Factor Analysis (FA)

Goal : find the latent variables within the observed ones

Latent representation of data (using SVD)

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^T = \mathbf{S}\mathbf{A}^T; \ \mathbf{U} \ is \ \mathrm{N} \times \mathrm{p} \text{ orthognal matrix } \mathbf{U}^{\mathbf{T}}\mathbf{U}=\mathrm{I}$$

where $\mathbf{S} = \sqrt{N}\mathbf{U}$ and $\mathbf{A}^T = \dfrac{\mathbf{D}\mathbf{V}^T}{\sqrt{N}}$

We have the following relations

$$\begin{cases} X_1 = a_{11}S_1 + a_{12}S_2 + \cdots + a_{1p}S_p \\ X_2 = a_{21}S_1 + a_{22}S_2 + \cdots + a_{2p}S_p \\ \vdots \\ X_p = a_{p1}S_1 + a_{p2}S_2 + \cdots + a_{pp}S_p \end{cases}$$

PCA estimates a latent variable model

$$\begin{cases} X_1 = a_{11}S_1 + a_{12}S_2 + \cdots + a_{1p}S_p \\ X_2 = a_{21}S_1 + a_{22}S_2 + \cdots + a_{2p}S_p \\ \vdots \\ X_p = a_{p1}S_1 + a_{p2}S_2 + \cdots + a_{pp}S_p \end{cases}$$

Due to hypothesis on X there are many solutions

$$\mathbf{X} = \mathbf{AS} = \mathbf{AR}^T\mathbf{RS} = \mathbf{A}^*\mathbf{S}^*$$

as

$$Cov(\mathbf{S}^*) = \mathbf{R}Cov(\mathbf{S})\mathbf{R}^T = \mathbf{I}$$

$\mathbf{S}^*$ is also the factor basis

Factor Analysis

Idea : use $q < p$

factor *loadings*

$$\begin{cases} X_1 = a_{11}S_1 + a_{12}S_2 + \cdots + a_{1q}S_q + \varepsilon_1 \\ X_2 = a_{21}S_1 + a_{22}S_2 + \cdots + a_{2q}S_q + \varepsilon_2 \\ \vdots \\ X_p = a_{p1}S_1 + a_{p2}S_2 + \cdots + a_{pq}S_q + \varepsilon_p \end{cases}$$

or $\mathbf{X} = \mathbf{AS} + \boldsymbol{\varepsilon}$, with $_j$ is mutually independent

Parameters are given by the covariance matrix

$$\boldsymbol{\Sigma} = \mathrm{AA}^T + \mathrm{D}_\varepsilon$$

with

$$\mathrm{D}_\varepsilon = \mathrm{diag}[\mathrm{var}(\varepsilon_1), \cdots, \mathrm{var}(\varepsilon_p)]$$

PCA and FA are related

FA can be further generalized

$$\begin{cases} X_1 = a_{11}S_1 + a_{12}S_2 + \cdots + a_{1q}S_q + \varepsilon_1 \\ X_2 = a_{21}S_1 + a_{22}S_2 + \cdots + a_{2q}S_q + \varepsilon_2 \\ \vdots \\ X_p = a_{p1}S_1 + a_{p2}S_2 + \cdots + a_{pq}S_q + \varepsilon_p \end{cases}$$

PCA : *all* variability in an item should be used
FA : only the variability in common is used

In most cases, the methods yield similar results PCA is preferred for data reduction
FA is preferred for structure detection

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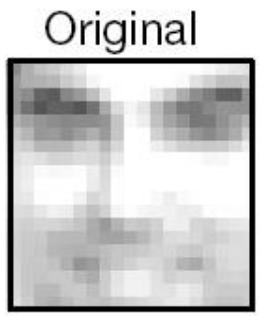
Problem

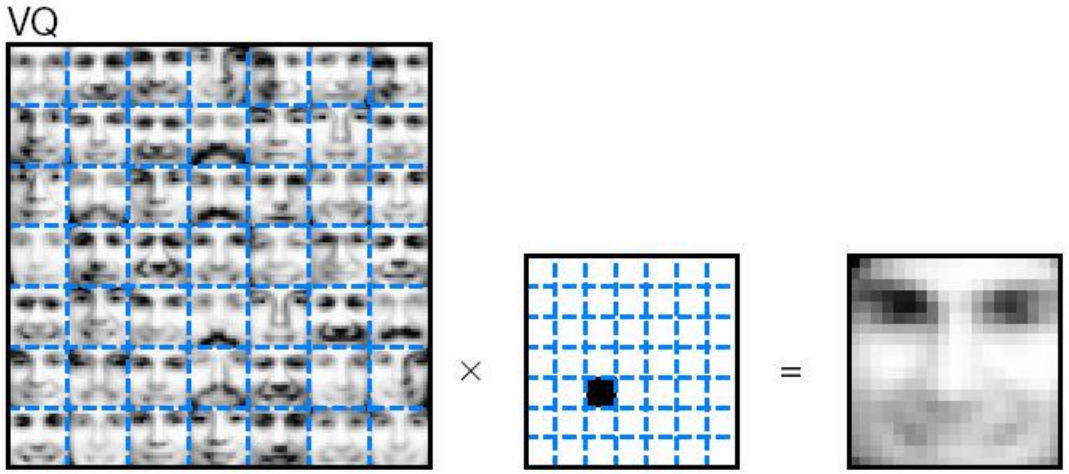
Given a set of images:

1. Create a set of basis images that can be linearly combined to create new images
2. Find the set of weights to reproduce every input image from the basis images
   - One set of weights for each input image

The reconstructed image is the basis image that is closest to the input image.

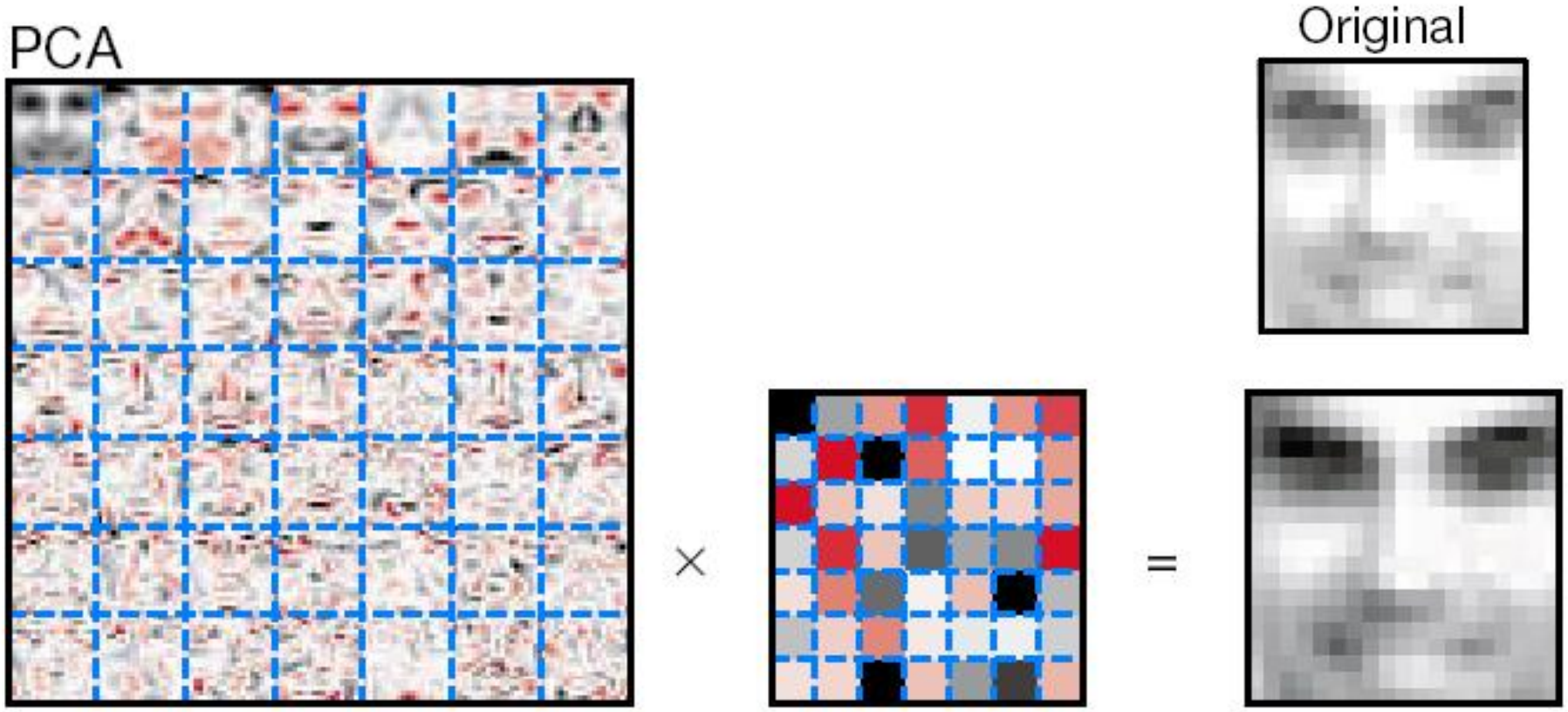
- Vector Quantization





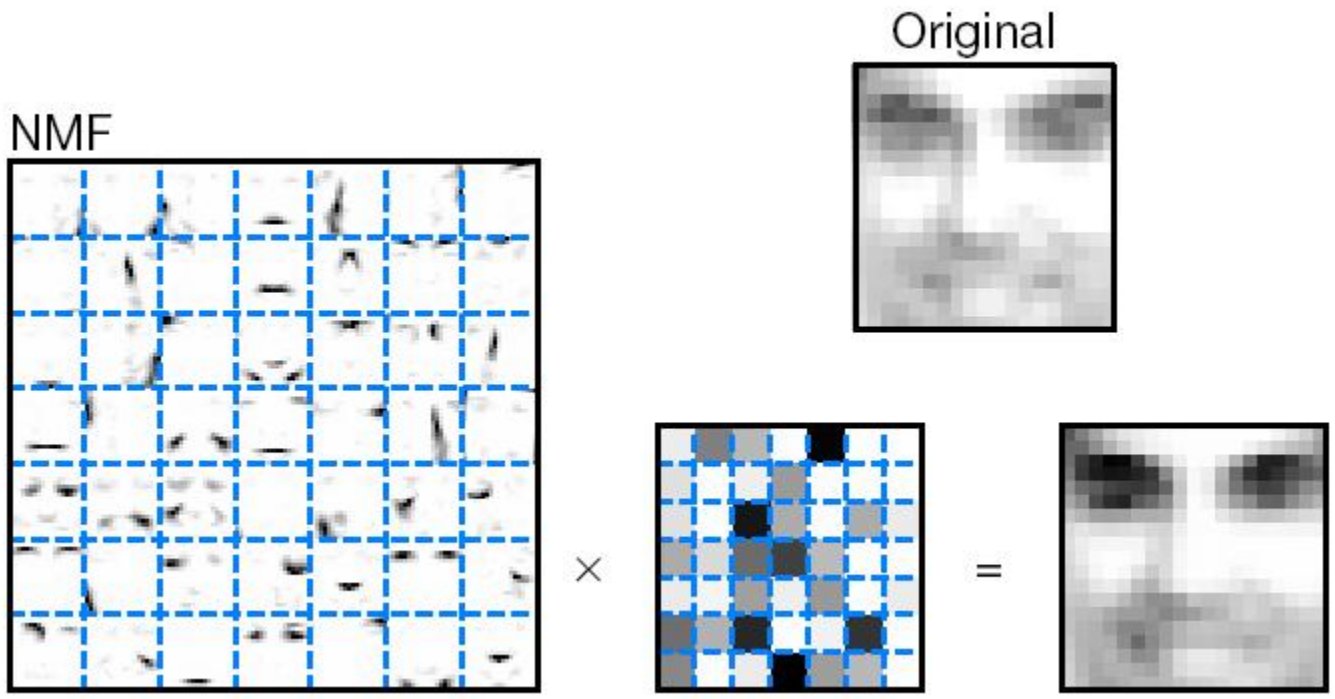
PCA

- Find a set of orthogonal basis images
- The reconstructed image is a linear combination of the basis images



Non-negative Matrix Factorization

- Like PCA, except the coefficients in the linear combination cannot be negative



- By constraining the weights, we can control how images are represented into the weighted sum of basis functions

Objective function

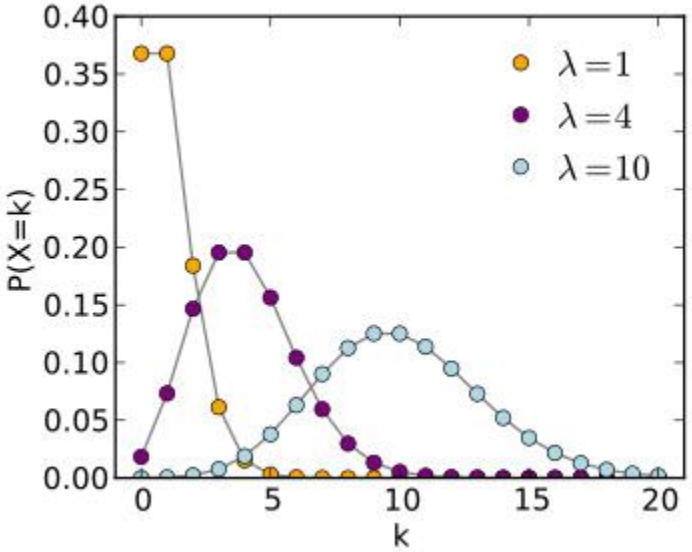
- Let the value of a pixel in an original input image be X.

$$X = WH + \varepsilon, \qquad W \in R^{N\times r}, H \in R^{r\times p}$$

- Let $(WH)_{ij} = \sum_k w_{ik} h_{kj}$ be the reconstructed pixel.
- If we consider X to be a noisy version of $(WH)_{ij}$, then the Poisson PDF of V is

$$P(x \mid (WH)_{ij}) = \frac{(WH)_{ij}^{x} e^{-(WH)_{ij}}}{x!}$$

$$P(x \mid \lambda) = \frac{\lambda^{x} e^{-\lambda}}{x!}$$



Objective function

$$P(x \mid (WH)_{ij}) = \frac{(WH)_{ij}^{x} e^{-(WH)_{ij}}}{x!}$$

- Now we will maximize the log probability of this PDF over W and H, leaving the relevant objective function to be:

$$L(W,H) = \sum_{i=1}^{N} \sum_{j}^{p} \left[x_{ij} \log(WH)_{ij} - (WH)_{ij} \right]$$

- Gradient ascent algorithm:

$$W = W + \eta \frac{\partial L(W,H)}{\partial W}; \qquad H = H + \eta \frac{\partial L(W,H)}{\partial H}$$

$$\frac{\partial L(W,H)}{\partial w_{ik}} = \sum_{j=1}^{p} [x_{ij} h_{kj} / (WH)_{ij} - h_{kj}];$$

$$w_{ik} \leftarrow w_{ik} + \eta \sum_{j=1}^{p} [x_{ij} h_{kj} / (WH)_{ij} - h_{kj}]$$

Objective function

- The objective function: $$L(W,H)=\sum_{i=1}^{N}\sum_{j}^{p}\left[x_{ij}\log(WH)_{ij}-(WH)_{ij}\right]$$

- Gradient ascent algorithm: $$w_{ik}\leftarrow w_{ik}+\eta\sum_{j=1}^{p}[x_{ij}h_{kj}/(WH)_{ij}-h_{kj}]$$

$$h_{kj}\leftarrow h_{kj}+\eta\sum_{i=1}^{N}[x_{ij}w_{ik}/(WH)_{ij}-w_{ik}]$$

- Question: How to keep Nonnegativity?
- Solution: Transfer the additive update algorithm to multiplicative ones

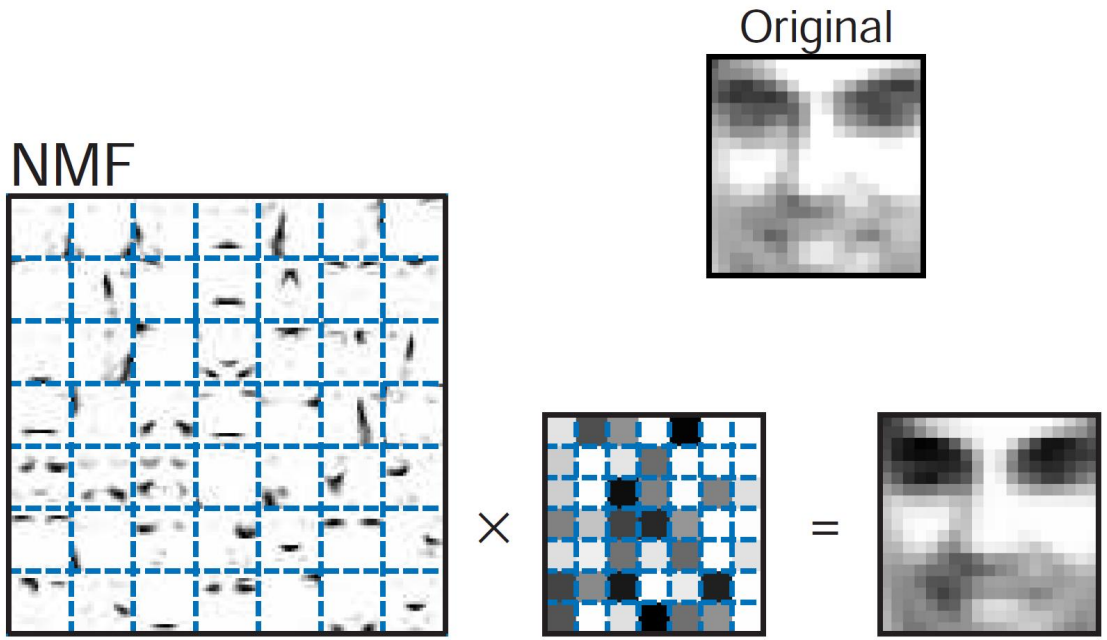
$$w_{ik}\leftarrow w_{ik}\frac{\sum_{j=1}^{p}h_{kj}\ x_{ij}/(WH)_{ij}}{\sum_{j=1}^{p}h_{kj}};\qquad h_{kj}\leftarrow h_{kj}\frac{\sum_{i=1}^{N}w_{ik}\ x_{ij}/(WH)_{ij}}{\sum_{i=1}^{N}w_{ik}}$$

Deriving Update Rules

- Gradient Descent Rule： (Alternating algorithm)

$$w_{ik} \leftarrow w_{ik} \frac{\sum_{j=1}^{p} h_{kj}\ x_{ij} / (WH)_{ij}}{\sum_{j=1}^{p} h_{kj}}; \qquad h_{kj} \leftarrow h_{kj} \frac{\sum_{i=1}^{N} w_{ik}\ x_{ij} / (WH)_{ij}}{\sum_{i=1}^{N} w_{ik}}$$

- Example: Face Basis



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